

Kushagra Omar

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ACADEMIC DETAILS

B Tech (E.C.E)	2022-2026	MMMUT, Gorakhpur	8.32/10.0 (CGPA)
Class XII (CBSE)	2022	B.S.S. Education Center	94.2%
Class X (ICSE)	2020	Methodist High School	93.67%

INTERNSHIP

AI Engineer Intern (On-Site) **AiProff.ai** **February 2026 – Present**

- Built an **Autonomous Procurement Negotiation Engine** using LangGraph, orchestrating a **5-agent AI system** to automate supplier analysis, should-cost modelling, and negotiation script generation end-to-end.
- Designed a **multi-agent architecture** with specialized agents for PPV Baseline, Should-Cost, BATNA/BCCS, Material Substitution, and Negotiation Strategy, using conditional routing, shared state propagation, and robust handling of partial data.
- Implemented a **graph-based Should-Cost model** using a directed acyclic graph (DAG) structure to represent hierarchical Bill-of-Materials (BOM) cost trees, enabling bottom-up cost rollups, node-level overrides, and transparent cost decomposition across raw materials, labour, and overhead components.
- Delivered a **production-ready AI platform** with FastAPI (REST + WebSocket) and React, featuring a tail-spend negotiation chatbot, human-in-the-loop supplier negotiation assistant, AWS SES automated negotiation emails, and a **PostgreSQL + Redis backend** on AWS EC2 with audit trails and procurement savings dashboards.

Project-Intern (On-Site) **Stealth and Camouflage Division, DRDO** **May 2025 – June 2025**

- Developed a robust **Python-based desktop application** using libraries like **Pandas, NumPy, Matplotlib and PyQt5** to visualize experimental trends and material performance metrics.
- Efficient import, filtering, and visualization of RCS experimental data.
- Implemented automated workflow to analyze key stealth parameters like Frequency vs. RCS, Frequency vs. RCSR, and RCS vs. Angle.
- Built high-resolution data visualization dashboard and correlation analysis to support material characterization.
- Performed correlation and trend analysis to identify material behaviors across varying frequencies and incident wave angles, improving clarity in stealth material performance evaluation.
- Collaborated with researchers to integrate data-driven findings into stealth material selection and optimization strategies.

PROJECTS

AI-Powered Multi-Agent PPT Generator | [\(Repository Link\)](#) | [\(Project Link\)](#)

- Built a **production-grade AI system** that converts PDFs, DOCX, and TXT files into **professional PowerPoint presentations** using **multi-agent** orchestration and RAG.
- Designed a **LangGraph-based multi-agent workflow** with dedicated agents for **outline generation, content expansion, quality review, and PPT export**, ensuring deterministic execution and shared state management.
- Implemented a **FAISS-backed RAG pipeline** with optimized chunking and similarity search to generate **context-aware, citation grounded slide content**.
- Developed a **dual-interface architecture** featuring a **Streamlit web UI** for users and a **FastAPI REST API** for scalable, programmatic access.
- Integrated **OpenAI LLMs and embeddings** with Pydantic schemas to produce **structured, hallucination-reduced content** and polished .pptx outputs via python-pptx.
- Technologies Used:** [Python](#), [LangGraph](#), [LangChain](#), [OpenAI LLM](#), [RAG](#), [FAISS](#), [FastAPI](#), [Streamlit](#), [Pydantic](#), [python-pptx](#)

ACHIEVEMENTS

- Among top 10 best performers in Hackathon-2024 – IEEE SB MMMUT
- Runner-up in Robokirti-2023 - Annual Technical Fest, MMMUT.
- Solved 150+ D.S.A. Problems on several platforms.
- Achieved 3rd position (in School) in Class 12th Board Exams, 2022.

SKILLS

Technical: Python, Java, LangChain, LangGraph, LangSmith, CrewAI, SciKit-Learn, PyTorch, MySQL, Git, GitHub, PostgreSQL, MongoDB, MLFlow, Data Version Control (DVC), DagsHub, Data Structures & Algorithm, Database Management System, OOPS, System Design.

Interests: Data Analysis, Machine Learning, Deep Learning, LLM’s, Generative AI, Multi AI-Agent Systems, NLP, Agentic AI, Image Processing, AI Automation, Retrieval Augmented Generation, MCP servers, Fine-Tuning of LLMs

Certificates: [DRDO Internship Certificate](#), [Long-Term Agentic Memory with LangGraph](#), [Getting Structured LLM Outputs](#), [LangChain for LLM Application Development](#), [Multi AI Agent System with crewAI](#)

Soft Skills: Problem solving approach, Analytical Thinking, Result Oriented, Flexibility, Trustworthiness, Ability to work in team environment.

POSITION OF RESPONSIBILITY

Senior Event Coordinator of Society of Automotive Engineers (SAE) **May 2025-ongoing**
Executive Member Society of Automotive Engineers (SAE) **July 2023 – May 2025**